



Interference effect on vortex-induced vibration in a parallel twin cable-stayed bridge



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ABSTRACT

A vortex-induced vibration (VIV), amplified by an interference effect caused by two parallel decks, was observed in the upstream deck of a twin cable stayed-bridge. This represents the first case of such an observation in an actual long-span cable-supported bridge. The observed VIV was successfully reproduced in a wind tunnel. Both decks were equipped with vanes and the single deck alone showed an allowable performance, in terms of VIV, based on wind tunnel tests that were carried out. It therefore appears that the observed vibration was affected by the parallel arrangement of the decks. A particle image velocimetry technique was successfully applied to investigate the complicated flow field between the upstream and downstream decks. Alternating eddies were formed in phase with the upstream deck motion and were transmitted to the downstream deck introducing “Ω” and “∩” shaped flow fields as a result. Following the alternating eddies, upward and downward wind streams were in turn fed into the gap and these flows amplified the vibration in the upstream deck of the bridge. Several modifications of aerodynamic additives were not effective in reducing this VIV. However, an increase in structural damping effectively mitigated the vibration.

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1. Introduction

The aerodynamic performance of long-span bridges is typically evaluated from the standpoints of both stability and serviceability. One of the important issues in serviceability performance is the mitigation of vortex-induced vibration (VIV), which is traditionally suppressed by applying several types of aerodynamic additives such as fairings or guide vanes. However, the majority of previous studies have focused on the VIV associated with single bridge decks.

Owing to the increase of traffic volume, an additional bridge could be added in parallel to an existing bridge. In this case, the second bridge is usually spaced with a minimal gap distance to increase economy and efficiency from the standpoint of road planning. Sometimes, two or more bridges are constructed in parallel from the initial design stage. For these types of parallel bridges, an interference effect on VIV is typically issued in the design stage for a slender box girder bridge (Honda et al., 1993) as well as small to medium sized cable-stayed bridges (Grillaud et al., 1992; Larsen et al., 2000a).

The planning of a parallel bridge, especially a long span cable-stayed bridge, requires that interference effects between adjacent structures be studied more intensively than usual. Kimura et al. (2008) reported that the interference effect could be significant even with a separation distance as large as 8 times the deck width in parallel cable-stayed bridges. Meng et al. (2011) proposed a strategy for mitigating the aerodynamic interference effect on VIV of two adjacent cable-stayed bridges. They applied a computational fluid dynamics approach to trace interference flow fields between two parallel decks. Both studies concluded that the interference effects were dependent on gap distance as well as deck shape. However the overall phenomena were quite complicated and more intensive studies were recommended to draw a general conclusion related to the interference effect on VIV.

Recently, VIV, magnified due to an interference effect caused by two parallel decks (hereafter referred to as interference VIV even though the parallel disposition may be one of the sources of the vibration), was observed in a twin cable-stayed bridge that is currently in service. The bridge was subjected to a steady wind of 9.8–11.5 m/s for two hours in the normal direction to the bridge axis. A lock-in heaving motion was monitored in the windward deck that exceeded the allowable limit for serviceability performance. Since the actual observation of the interference VIV in a parallel cable-stayed bridge that was in service was the first such case, a series of investigations were carried out in a wind tunnel

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in an attempt to identify the main sources of the vibration as well as possible measures that could be used for mitigation. Particle image velocimetry (PIV) tests were utilized to investigate the complicated flow field between the two decks.

2. Field observation of interference VIV

2.1. The bridge investigated

The Jindo Bridge, shown in Fig. 1, is composed of two parallel cable-stayed bridges with a main span length of 340 m. The 1st Jindo Bridge shown in the left side of Fig. 1 was opened to traffic in 1984 while the 2nd Jindo Bridge on the right side of Fig. 1 was built in 2005. These bridges are hereafter referred to as Bridge 1 (older bridge) and Bridge 2 (newer bridge). The decks of the two bridges are similar in cross-sectional shape as shown in Fig. 2, and closely spaced with a net distance of 10 m between them, as shown in Fig. 3. Bridge 1 was equipped with vanes to mitigate conventional VIV from the beginning of the bridge design.

The vanes are different with those adopted for the Storebælt suspension bridge (Larsen et al., 2000b) in terms of the shape and position. Based on the configuration of the vane, two different contributions would be expected for mitigating VIV. One function is as a fairing which modifies the corner shape with a rounded curve and the other function as an actual guide vane with a provided gap. However, each contribution is not confirmed in the present study and the attachments are simply referred to as vanes according to the design report (Hyundai Engineering and Construction (HDEC), 2000a).

2.2. Identified natural frequencies of the bridges

An international test-bed application of a smart wireless sensor network has been ongoing for Bridge 2. The sensor network consists of a total of 113 sensors and the natural frequencies of Bridge 2 were identified as shown in Table 1 (Jang et al., 2010; Spencer et al., 2011). The natural frequencies of Bridge 1 were identified from a built-in monitoring system and are also included in Table 1 (Hyundai Engineering and Construction (HDEC), 2000b). The fundamental vertical frequencies of both bridges are similar but the frequency of Bridge 2 (0.45 Hz) is slightly lower than that of Bridge 1 (0.513 Hz).

2.3. The observed interference VIV

On April 19, 2011, a significant vibration was observed on the deck of Bridge 2 for a duration of two hours. The onsite wind velocity and the vibration of the deck were recorded with the built-in monitoring system installed in two bridges, as shown

in Fig. 4. The wind velocity was measured by the ultrasonic anemometer installed on one side of the deck of Bridge 1 since the anemometer on Bridge 2 was temporarily unavailable. The anemometer was placed at a height of 3 m from the deck and the maximum sampling frequency was 10 Hz. The response of the deck was measured by accelerometers installed on both sides of the deck at the center of main span. The displacement at the same position was also monitored by a set of the laser displacement transducer located on the crossbeam of a pylon and the target

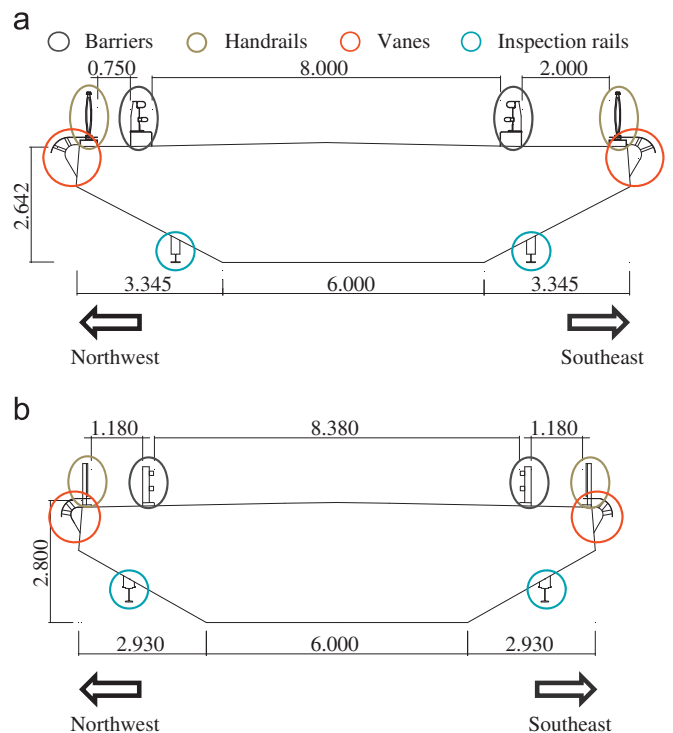


Fig. 2. Cross sections of (a) Bridge 2 (newer) and (b) Bridge 1 (older) (unit: m).

Table 1 Identified natural frequencies (Hz) of bridges.		
Modes	Bridge 2 (Jang et al., 2010; Spencer et al., 2011)	Bridge 1 (Hyundai Engineering and Construction (HDEC), 2000b)
Vertical (first)	0.45	0.513
Vertical (second)	0.66	–
Vertical (third)	1.03	–
Torsional (first)	1.87	1.490



Fig. 1. Parallel twin Jindo Bridge.

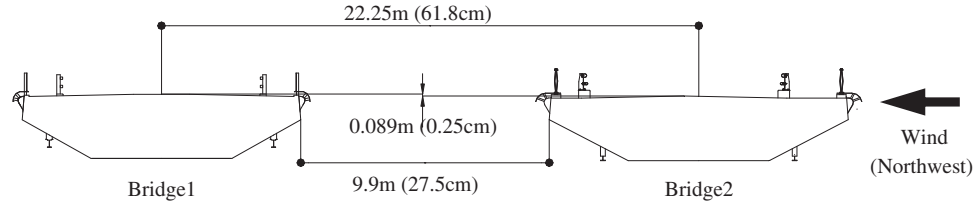


Fig. 3. Disposition of the two bridge decks with wind directions (Dimensions for the scaled model are shown in parentheses.).

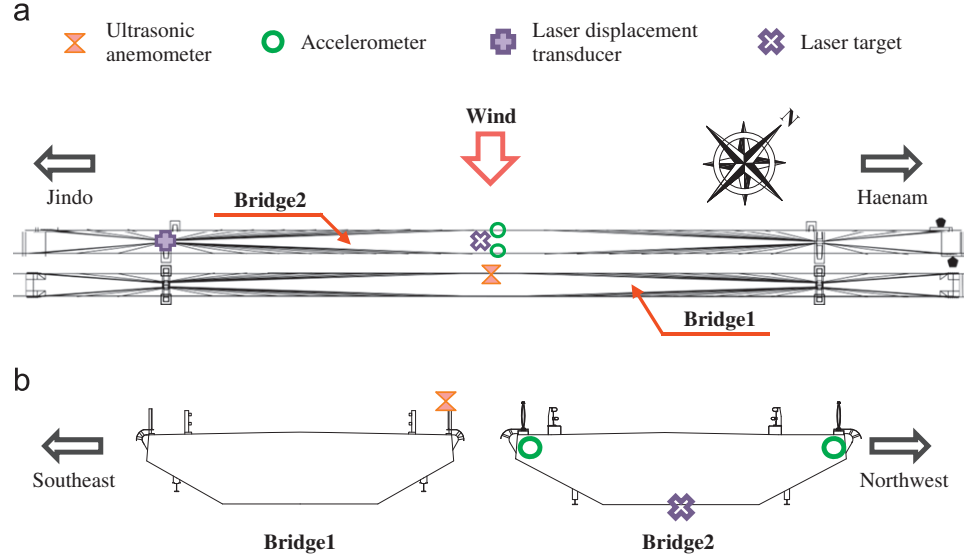


Fig. 4. Built-in monitoring sensors used for the observation of interference VIV: (a) plane view and (b) cross section.

attached on the bottom of the deck at the center of main span. Fig. 5 shows the time histories of wind velocity, displacement and acceleration at the center of the main span. The sampling frequency of the data logger was set to 100 Hz.

Fig. 6 shows the frequency component of the displacement time history and demonstrates the deck was excited with a single dominant frequency of 0.438 Hz which is the first vertical frequency of Bridge 2. The detrended displacement was multiplied by the square of the circular frequency of the first vertical mode and compared to the time history of acceleration so as to confirm general coincidence between both signals. The maximum single amplitude of displacement is estimated to be 0.188 m after eliminating temporal effects associated with ongoing vehicle traffic during the observation.

The wind velocity time history in Fig. 5(a) includes several spikes which seems to be originated from the malfunctioning of the ultrasonic anemometer. However, the number of spikes was somehow limited and the 10-minute averaged wind velocity was estimated by a moving average concept as shown in Fig. 5(a) with a red line.

According to a design guideline for steel cable-supported bridges (Korean Society of Civil Engineers (KSCE), 2006), the maximum allowable acceleration is recommended to be less than 0.5 m/s^2 up to a wind velocity of 25 m/s at a deck level to satisfy the serviceability performance. The measured acceleration in Fig. 5(c) clearly exceeded this allowable limit. The 10-minute averaged wind velocity was estimated to be 9.8–11.5 m/s during the time when acceleration exceeded the allowable range.

The Korean peninsula is subjected to monsoon winds from the continental Asia to the Pacific Ocean during the winter and spring seasons (Korea Meteorological Administration (KMA), 2012). The observed wind also blew from the northwest and Bridge 2 was

located upstream thus providing a normal disposition to the arriving wind, as illustrated in Fig. 7.

3. Reproduction of interference VIV in a wind tunnel

3.1. Test setup

The observed interference VIV was reproduced in a wind tunnel operated by the Department of Civil and Environmental Engineering at Seoul National University. The test section of the wind tunnel is 1.5 m in height, 1.0 m in width and 4.0 m in length. The section models for both bridge decks were prepared with the length scale of $\lambda_L = L_m/L_p = 1/36$, where L_m and L_p represent the length of model and prototype, respectively. The frequency scale $\lambda_f = f_m/f_p = 1/\lambda_T = 5$ is arbitrarily determined without considering the similarity of Froude number, which is a general practice in a section model test. Here λ_T is the time scale and f_m and f_p represent the frequency of the model and the prototype, respectively. The scaling of reduced velocity is satisfied as follows.

$$\left(\frac{U}{fB}\right)_m = \frac{\lambda_L/\lambda_T}{\lambda_f\lambda_L} \left(\frac{U}{fB}\right)_p = \left(\frac{U}{fB}\right)_p \quad (1)$$

where U is the wind velocity and the subscripts m and p denote the model and the prototype, respectively. Table 2 shows the setup parameters for both section models on spring-supported systems.

The experimental setup for the wind tunnel test is shown in Fig. 8 with the sensor deployment. Both decks were equipped with end plates of $0.40 \text{ m}(\text{width}) \times 0.13 \text{ m}(\text{height})$. The section models were made of balsa and were colored black for PIV tests.

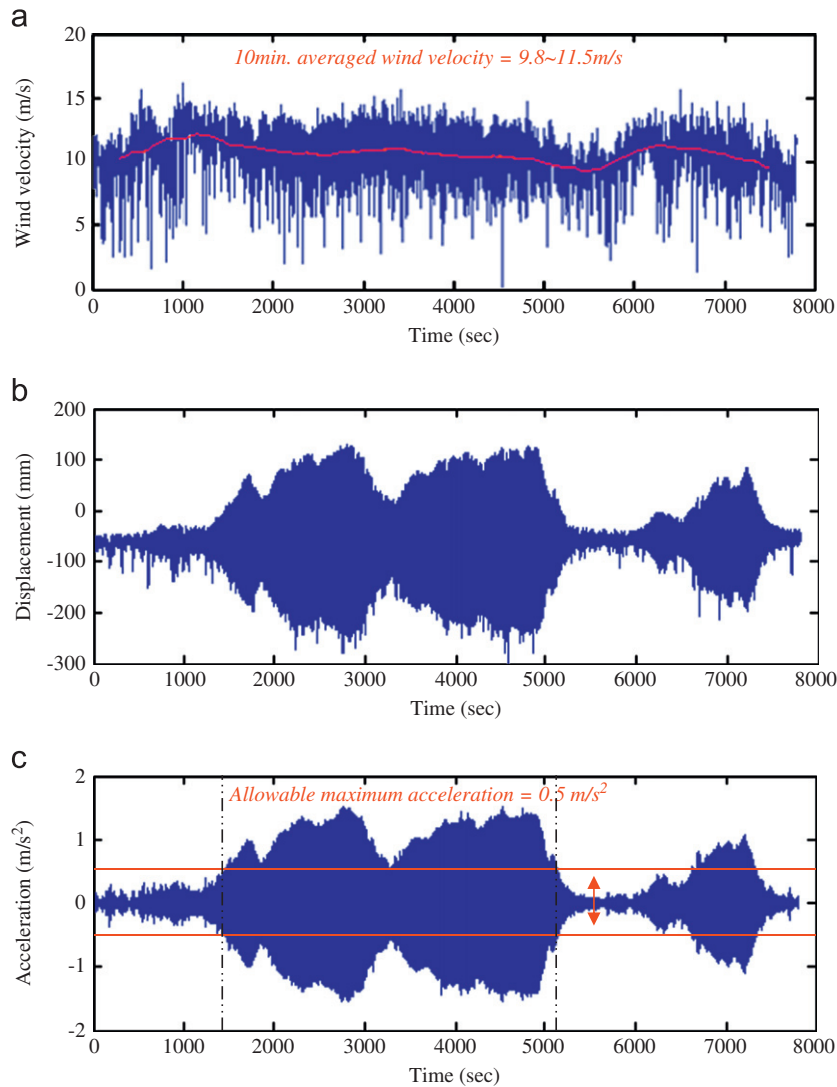


Fig. 5. Observed wind velocity and interference VIV at the center of the main span in Bridge 2: (a) wind velocity, (b) displacement and (c) acceleration.

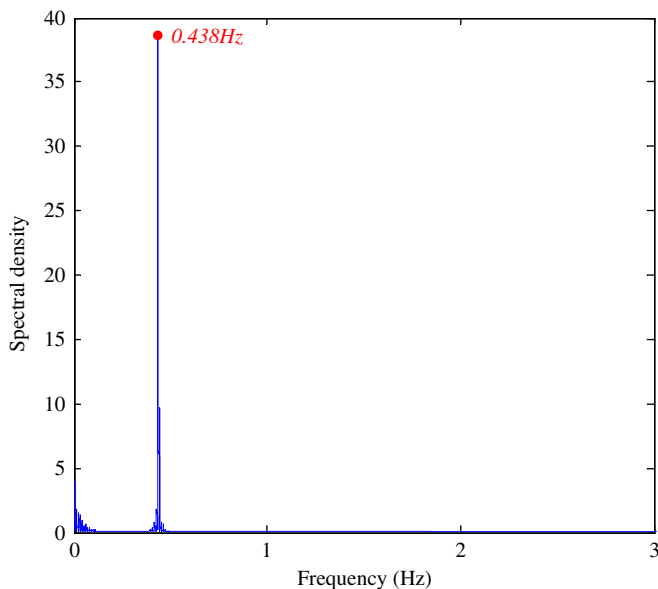


Fig. 6. Single-sided spectrum of the measured time history of displacement.

In order to observe flow field near the section model, the end plates were made of transparent acrylic plastic. Total 8 (4 for each section model) laser displacement transducers were used for measuring vertical and torsional displacements of both decks with a sampling frequency of 400 Hz.

The design guideline (Korean Society of Civil Engineers (KSCE), 2006) recommends a 0.40% damping ratio for the aerodynamic design of a steel deck cable-stayed bridge with welded connections. To better observe VIV, however, the damping ratios were set to a minimal level of 0.10% for both the vertical and torsional modes. This level of damping is hereafter referred to as the low damping ratio. To examine the bridge motions for the design level damping ratio, the damping ratios were also set to a proper level of 0.40% for the vertical mode and 0.39% for the torsional mode. This level of damping is hereafter referred to as the design damping ratio.

3.2. VIV of the single Bridge 2

Prior to investigating interference VIV, the aeroelastic behavior of the single Bridge 2 was evaluated with the conditions for Bridge 1 being excluded. Fig. 9 shows the velocity (of wind)–amplitude (of response) curve (V – A curve) for the vertical and

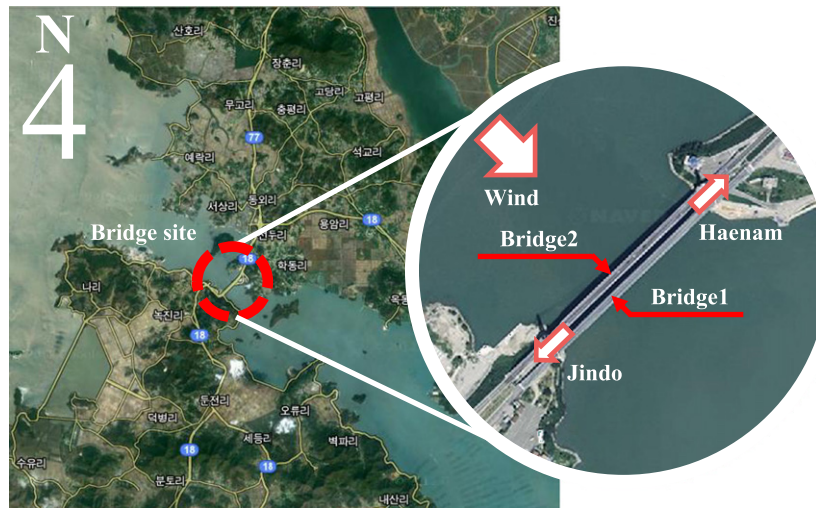


Fig. 7. Disposition of the bridges.

Table 2
Similarity scales and setup parameters for sectional model wind tunnel tests.

Parameters	Similarity scale	Bridge 2			Bridge 1		
		Prototype	Model (target)	Model (measured)	Prototype	Model (target)	Model (measured)
Length (m)	$\lambda_L = 1/36$	32.400	0.900	0.900	32.400	0.900	0.900
Width (m)	λ_L	12.690	0.353	0.353	11.860	0.353	0.329
Height (m)	λ_L	2.750	0.076	0.076	2.800	0.078	0.078
Mass (kg/m)	λ_L^2	8978	6.927	6.697	6950	5.363	5.304
Mass moment of inertia (kg m ² /m)	λ_L^4	152,836	0.091	0.090	84,030	0.050	0.052
1st Vertical frequency (Hz)	$\lambda_f = 5$	0.436	2.180	2.180	0.513	2.561	2.550
1st Torsional Frequency (Hz)	λ_f	1.834	9.165	8.867	1.490	7.444	7.533
Frequency Ratio	1	4.204	4.204	4.067	2.906	2.906	2.954
Vertical Damping Ratio (%)	1	–	–	0.100	–	–	0.100
Torsional Damping ratio (%)	1	–	–	0.100	–	–	0.100

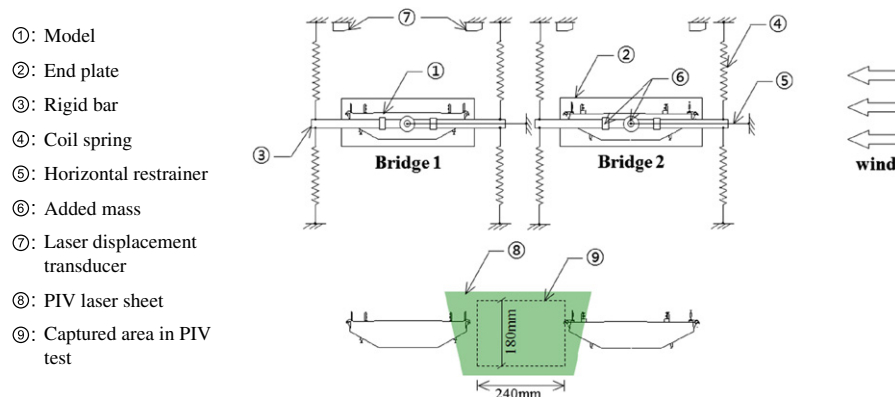


Fig. 8. Setup of wind tunnel tests.

torsional displacements of the single Bridge 2. In the Fig. 9, ζ_h and ζ_α represent the damping ratios for the vertical and torsional modes, respectively. Note that all vibrational responses are given in single-amplitude formats in this study.

It would be preferable to express the parameters of wind tunnel test such as the wind velocity and section model responses in terms of non-dimensional values for the identical representation between a prototype bridge and a wind tunnel model. However,

since the intent of this study was to investigate the field monitored responses and the wind tunnel tests were carried out for the quantitative reproduction of the observed vibration, all the parameters regarding the wind tunnel tests are represented in prototype scales by multiplying the scale ratios defined in Table 2 and Eq. (1).

For the case of a low damping ratio, the original section with the as-is vanes showed a conventional VIV for heaving and

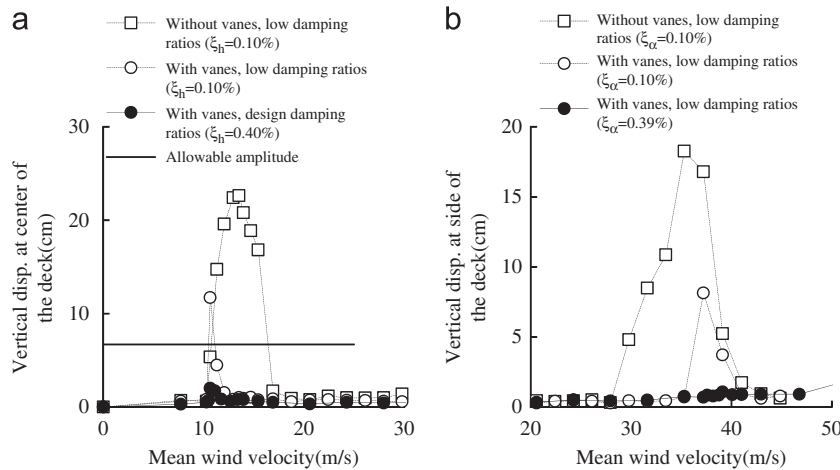


Fig. 9. V–A curves for Bridge 2 alone at the center of the main span with and without the as-is vanes: (a) vertical displacement and (b) torsional rotation.

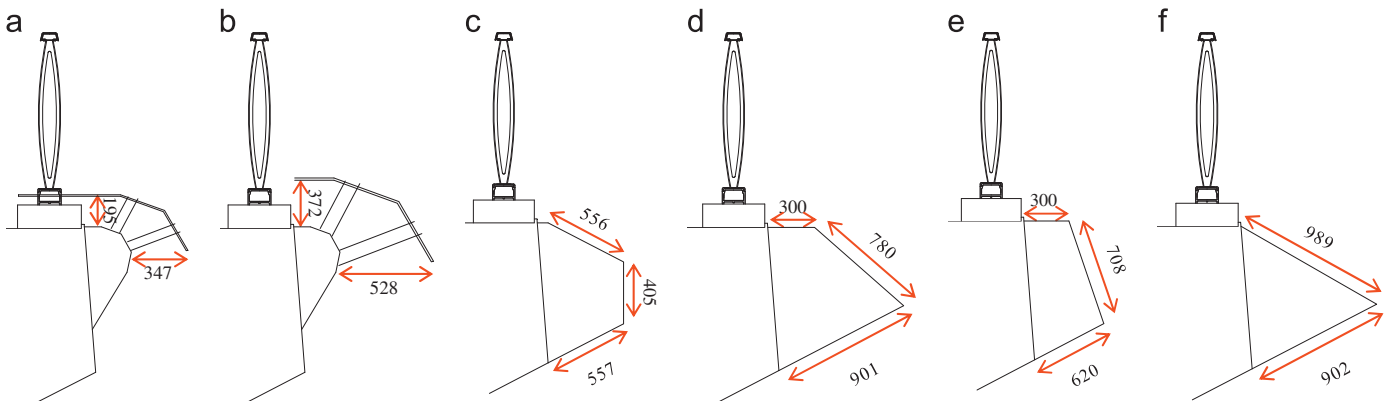


Fig. 10. As-is vane and alternative aerodynamic attachments: (a) as-is vanes, (b) enlarged vanes, (c) fairing 1, (d) fairing 2, (e) fairing 3 and (f) fairing 4 (unit: mm).

torsional motions, as shown in Fig. 9, and the amplitude exceeded the allowable limit specified in the design guideline (Korean Society of Civil Engineers (KSCE), 2006). The allowable limit is specified with the vertical acceleration of 0.5 m/s^2 and can be converted to a vertical displacement of 0.066 m considering a single harmonic motion with the first vertical mode. However, the maximum amplitude of the heaving and torsional motions decreased below the allowable limit for the case of design damping ratio. Both V–A curves were obtained under laminar wind conditions with a turbulence intensity of less than 1%. For purposes of comparison, Fig. 9 also shows the V–A curves with the as-is vanes eliminated from the single deck. As shown in the figure, both heaving and torsional motions show large amplitudes and this illustrates the contribution of as-is vanes in mitigating the VIV of a single deck.

Several alternatives to the as-is vanes were also tested to determine whether any possible enhancement in aerodynamic additives might be effective in reducing the heaving VIV for the single Bridge 2. As shown in Fig. 10, an enlarged vane was also tested. Even though the fairing in Fig. 10(f) showed the best performance among the alternatives considered, the overall performances were not better than that when the as-is vane was used. This will be demonstrated in Section 3.3. However, a wider range of possible fairing shapes was not considered in the current investigation.

3.3. Interference VIV of Bridge 2 with fixed Bridge 1

During the vibration of Bridge 2, no detectable vibration was found for Bridge 1. Accordingly, the reproduction of the vibration

of Bridge 2 was performed with the fixed setting of Bridge 1, which simplified the experimental setup without significant loss of generality.

Fig. 11 shows the increase in the VIV amplitude due to the parallel disposition of the two bridges. Bridge 2 was located upstream as was the case of the field observation. The increased vibration reached up to 34 cm in single amplitude for the low damping case. However, the setting with a design damping ratio decreased the amplitude below the allowable limit. For purposes of comparison, the observed vibration in field is also included in Fig. 11.

To compare the amplitudes between the interference VIV for the two parallel bridges and the conventional VIV for the single bridge, both V–A curves are drawn together in Fig. 12(a). Fig. 12(a) clearly shows that interference VIV amplifies the vibration in the upstream Bridge 2. The fairing case (fairing 4 in Fig. 10) is also shown in Fig. 12(b) to estimate any possible contribution of the alternative fairing in reducing interference VIV. The alternative fairing case seems to provide almost equal or slightly worse performances compared to the case of as-is vanes.

It was also found that the onset wind velocity for the interference VIV increased slightly from the onset velocity of VIV for a single bridge. The origin of this phenomenon is from the change in the Strouhal number for the upstream section owing to the parallel arrangements of the decks. The Strouhal number of the Bridge 2 was measured to be 0.109 for the case of a parallel arrangement while it was 0.123 for the case of a single deck.

Consequently, the amplification of the VIV in the upstream deck due to interference effect could be confirmed through wind

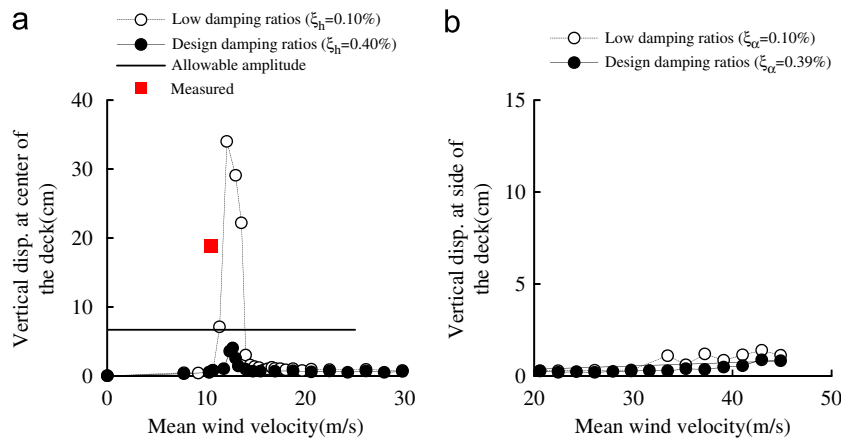


Fig. 11. V-A curves for parallel Bridge 2 (upstream) with as-is vanes: (a) vertical displacement and (b) torsional rotation.

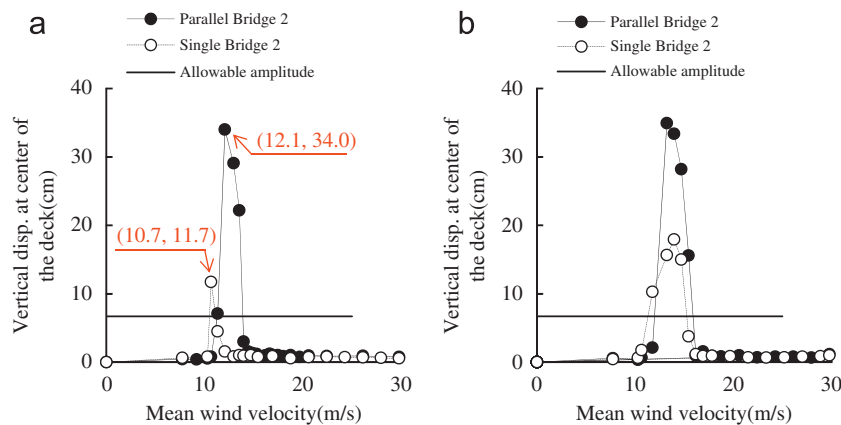


Fig. 12. Comparison of vertical VIV in Bridge 2 for the cases of parallel disposition of two bridges and single bridge alone with low damping ratio ($\xi_h=0.10\%$, $\xi_\alpha=0.10\%$): (a) as-is vanes and (b) alternative fairing 4.

tunnel tests. The modification of cross sections with other types of aerodynamic additives does not appear to be effective in terms of mitigating the interference VIV for the bridge within the investigated ranges of this study.

3.4. Mitigation of interference VIV with increase of damping ratios

As shown in Fig. 11, the amplitude of the interference VIV was influenced by the mechanical damping ratio of the bridge structure. Accordingly, the damping ratio of Bridge 2 in a heaving mode was varied during setting of the section model and the corresponding change in the maximum amplitude was examined. Viscous damping of the structure was simulated with several types of balls and plates that were immersed in oil. Since the damping ratio is dependent on the amplitude of the vibration, the consistent part of the decaying free vibration signals was utilized to identify the damping ratios as shown in Fig. 13. The damping ratio for the heaving motion was varied from 0.10% to 0.9% to the critical damping ratio and the corresponding change in amplitude was obtained as shown in Fig. 14. The interference VIV decreased rapidly with increasing damping ratio of the structure. From the figure, it is estimated that a minimum damping ratio of 0.4% is necessary to mitigate the interference VIV below the allowable limit.

Since the damping ratio is a critical factor for evaluating performance, the damping ratio for the first vertical mode of Bridge 2 was estimated from ambient vibration data obtained from the built-in accelerometers shown in Fig. 4. The vertical responses were continuously accumulated from 10 a.m. on

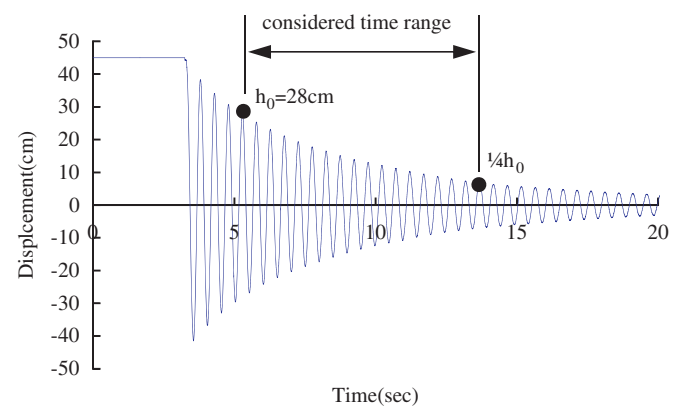


Fig. 13. The region of the decaying time history used for the evaluation of damping ratio.

November 15 to 7 p.m. on November 17, 2012. The overall averaged wind velocity was determined to be 4.7 m/s during that period. Most of the vibration was induced by vehicle traffic over the bridge. The acceleration data were divided by a 1-hour unit as shown in Fig. 15(a). As a result, a total of 57 time history sets were separately utilized for the identification of structural damping ratio of Bridge 2. The average value of maximum accelerations in each data set was determined to be 0.238 m/s^2 . The damping ratios for each data set were estimated by the Natural Excitation

Technique (NExT) paired with the Eigensystem Realization Algorithm (NExT-ERA) (James et al., 1993; Juang and Phan, 2001; Siringoringo and Fujino, 2008). Fig. 15(b) shows the distribution

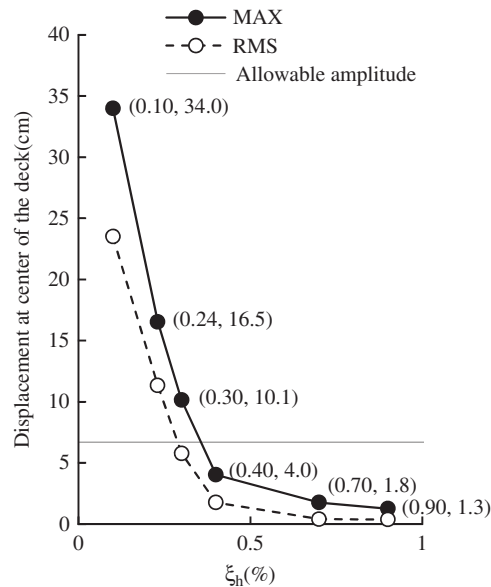


Fig. 14. Mitigating the interference VIV of Bridge 2 by increasing the damping ratio.

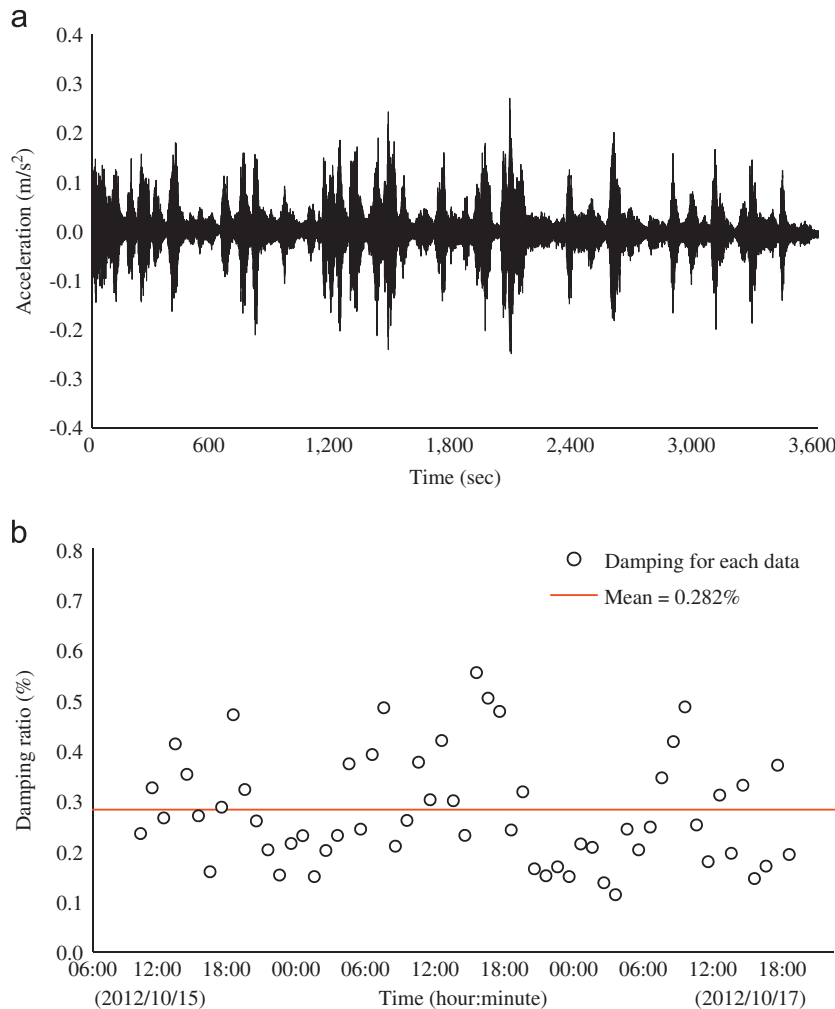
of the estimated damping ratios for each 1-hour unit data. Even though some scattering was observed, the mean value of identified damping ratios was estimated to be 0.28%. The median and mode of the estimated damping ratios were slightly lower at 0.25%. This unexpected low level of mechanical damping ratio possibly accounts for the observed vibration.

4. Flow field observation

4.1. Test setup

The PIV tests were utilized to observe the flow fields behind Bridge 2. While the reproduction of the interface VIV was performed with the fixed Bridge 1, the PIV tests were performed under movable conditions for both decks with both the bridge sections being installed on each spring-support system as shown in Fig. 8. For purposes of comparison, the PIV tests were also performed for the case of single Bridge 2.

The setting parameters followed Table 2. The flow field was captured by a high resolution, two-mega pixel CCD camera. The captured images cover a sufficient area between the two deck sections as illustrated in Fig. 8. For the case of single disposition of Bridge 2, the camera captured the same-size images as for the case of parallel disposition. A laser device supplied a 532 nm laser with a light intensity of 135 mJ. The laser device had two heads



which generated pulses at time intervals of 500 μ s. Very fine olive oil particles of 1 μ m were used for seeds.

The wind velocity was set to 11.6 m/s at which Bridge 2 showed a large amplitude for the interference VIV. For the case of single

Bridge 2, the wind velocity was set to 11.0 m/s which showed also a large amplitude for the VIV. The flow fields at almost equally spaced 1/16 phases of one entire period were monitored during the steady-state lock-in heaving motion. The displacement signal obtained

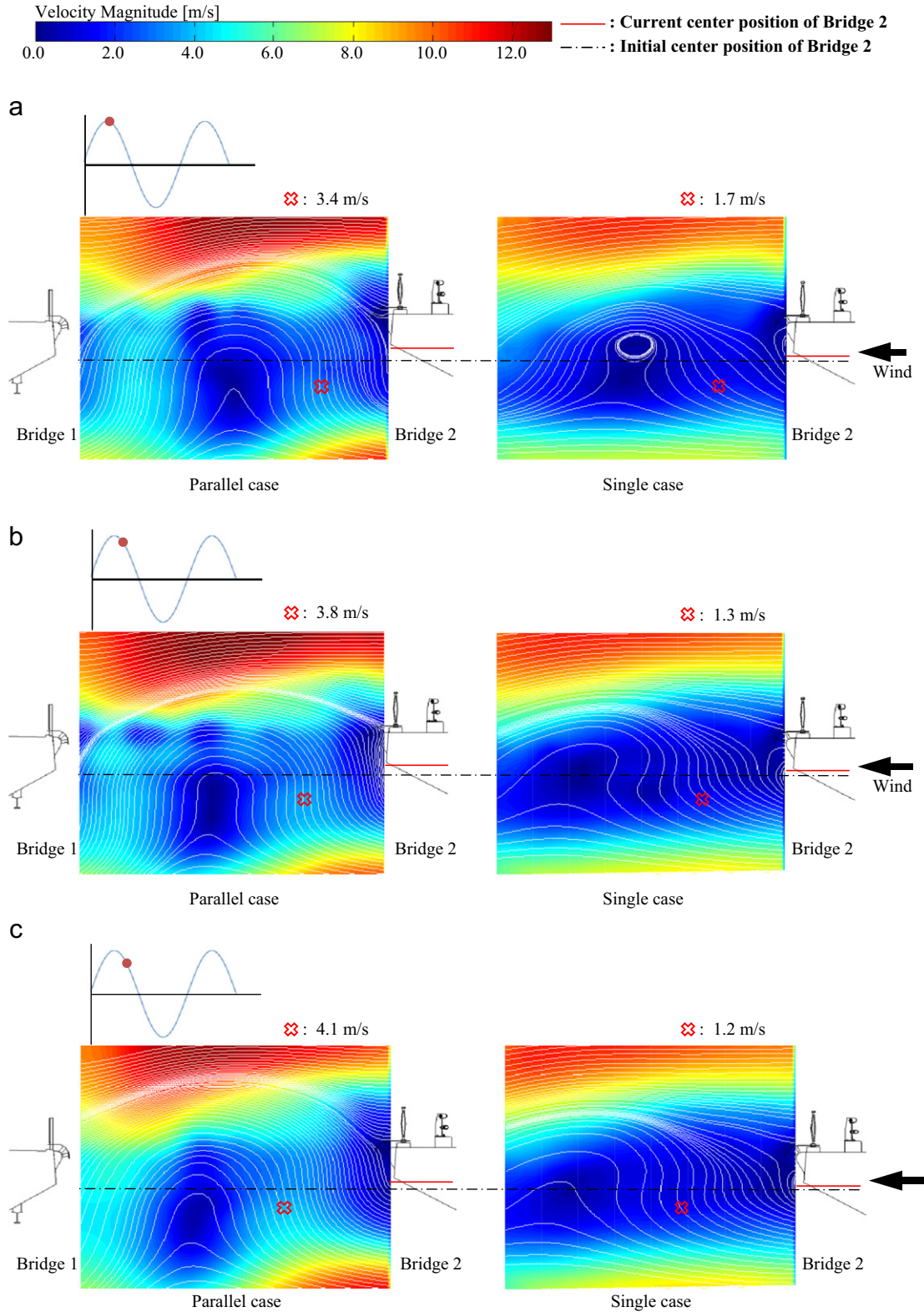


Fig. 16. Wind velocity fields and streamlines during one period of heaving motion in Bridge 2 at phases (a) 0, (b) $\pi/8$, (c) $\pi/4$, (d) $3\pi/8$, (e) $\pi/2$, (f) $5\pi/8$, (g) $3\pi/4$, (h) $7\pi/8$, (i) π , (j) $9\pi/8$, (k) $5\pi/4$, (l) $11\pi/8$, (m) $3\pi/2$, (n) $13\pi/8$, (o) $7\pi/4$ and (p) $15\pi/8$.

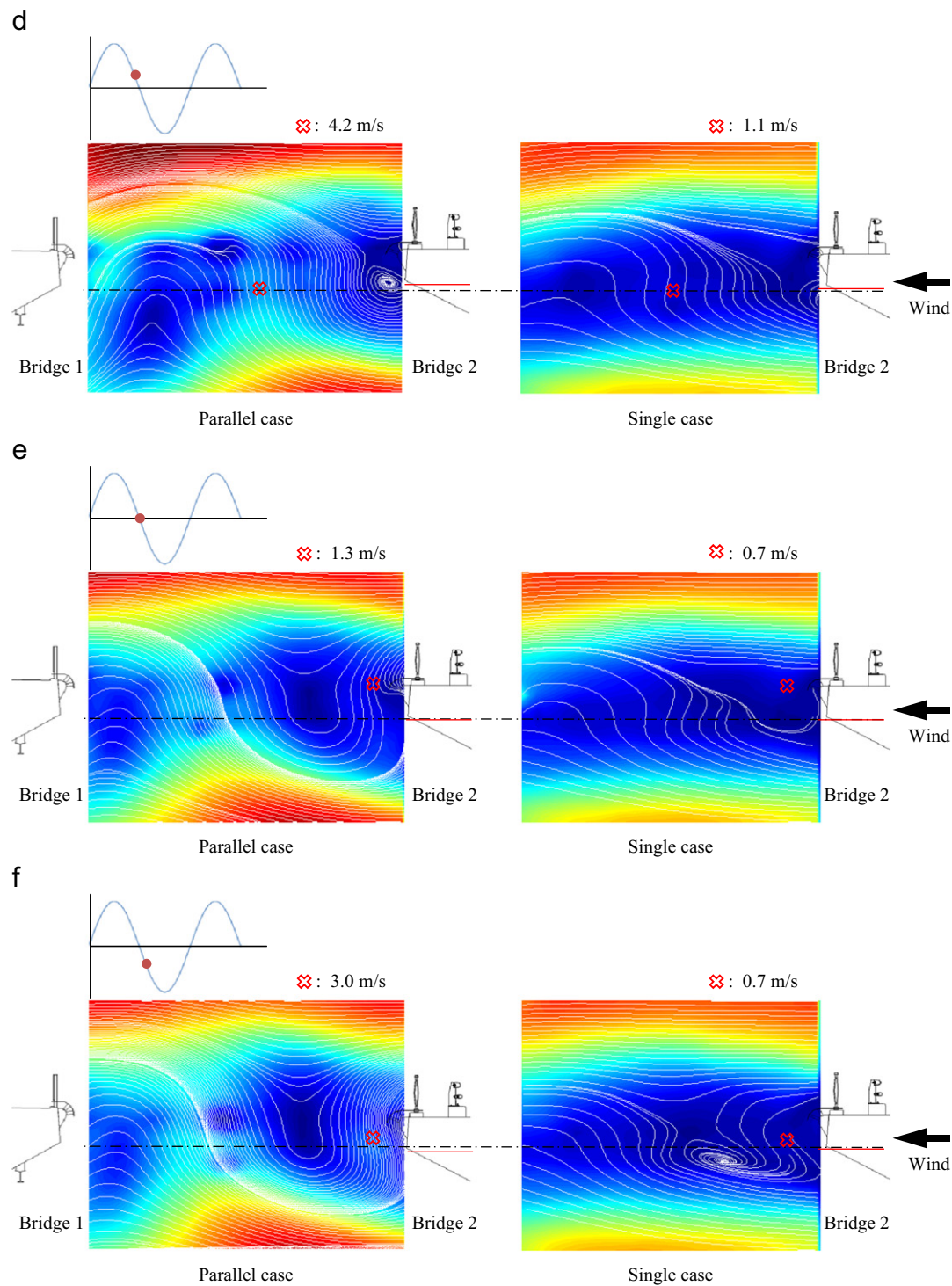


Fig. 16. Continued.

from the laser displacement transducer for Bridge 2 was amplified and utilized as a trigger of PIV capture. The position of the deck at the instance of PIV capture was controlled by the time delay between the peak of the amplified displacement and the target phase to capture. The repeated PIV pictures at one phase produced very similar flow patterns. Accordingly, 500 repeated images were saved and averaged at a specific phase to obtain a clear picture of the flow pattern (Kim, 2002; Hussain and Reynolds, 1970).

4.2. Flow fields at each phase

Fig. 16 shows the change in flow patterns for the consecutive increments in phase for an entire period of oscillation. The left-hand side figures show the case of parallel disposition while the right-hand side figures illustrate the case of the single Bridge 2. The velocity contours show that the wind speeds behind Bridge 2 in both cases are relatively low.

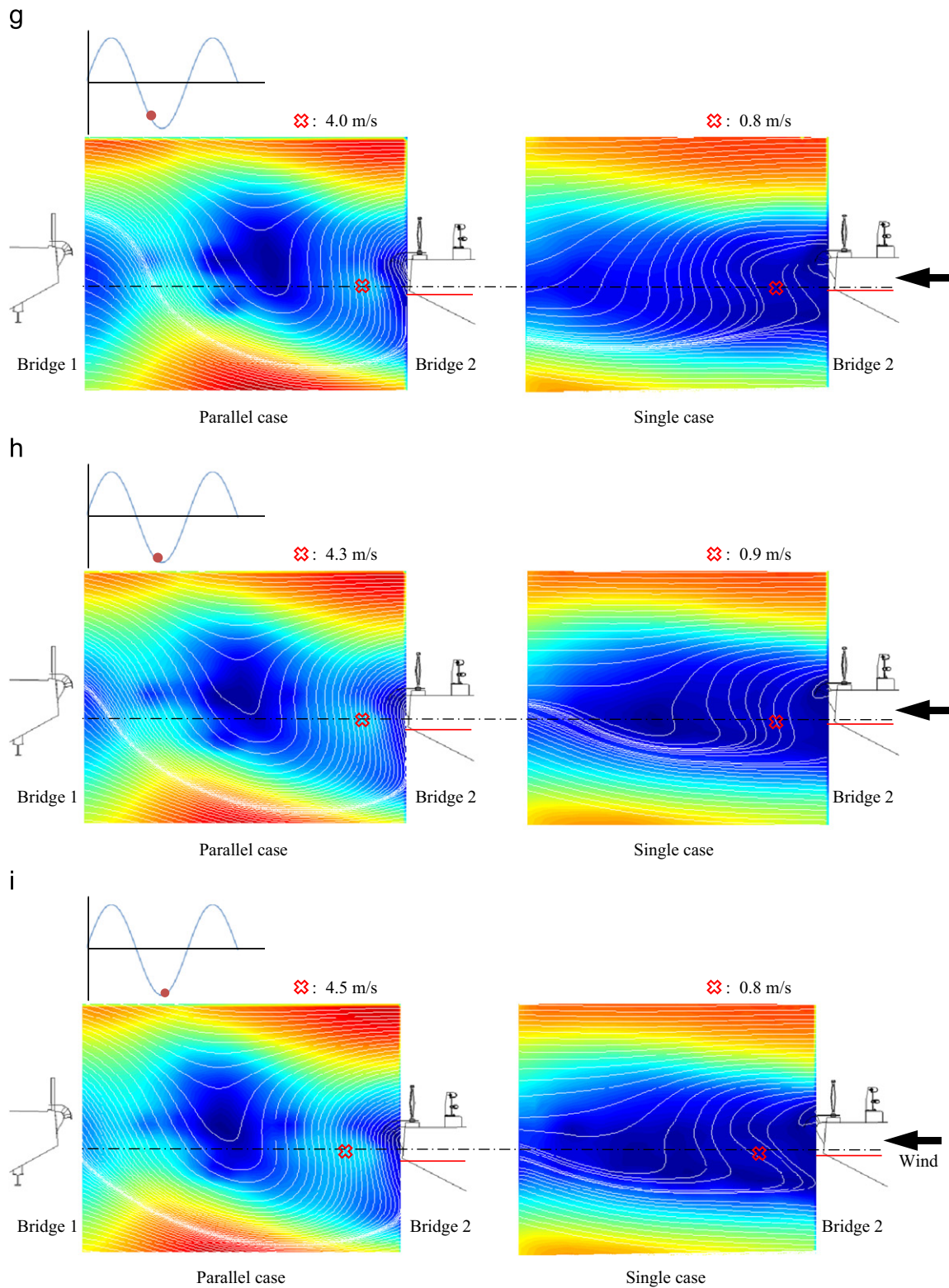


Fig. 16. Continued.

The variation in flow field for the case of parallel disposition is discussed first. At the top position of Bridge 2 (Fig. 16(a)), the flow makes a “Ω” shape in which the wind flow enters from the bottom side and moves upward, resulting in an upward force on Bridge 2. However, the flow turns down and passes under Bridge 1. As Bridge 2 moves back to the neutral position, a clockwise eddy is initiated at the bottom of the deck (Fig. 16(d)). This eddy travels down to Bridge 1 making a “U” shape streamline and the overall flow

pattern changes to a downward direction behind Bridge 2. However, the flow turns up and passes over Bridge 1 (Fig. 16(f)–(h)). As Bridge 2 moves back to the neutral position again after reaching the bottom point, the counterclockwise eddy is initiated at the top of the deck section (Fig. 16(k)–(m)). This eddy travels down to Bridge 1 making a “Ω” shaped flow field (Fig. 16(n)–(p)). Consequently, the consecutive PIV images for the case of the parallel disposition shows a clear change of flow pattern between the “Ω” and “U” shapes alternating

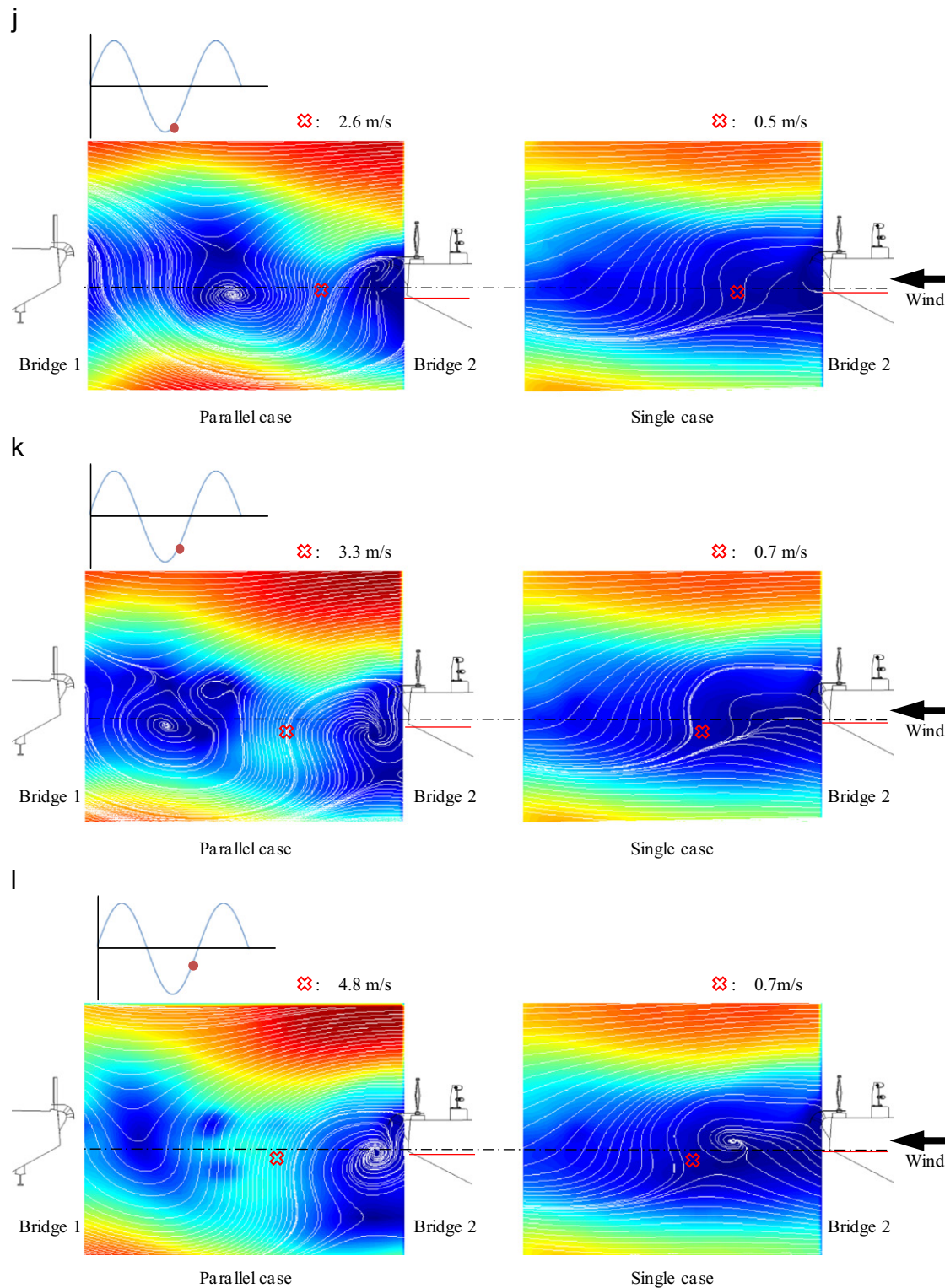


Fig. 16. Continued.

the streamlines upward and downward behind Bridge 2. The wind velocity of the streams between two consecutive eddies is in the range of 3.6–5.0 m/s as indicated in Fig. 16.

The PIV images for the case of single Bridge 2 are also shown in Fig. 16 for the same phase angle even though the amplitude is smaller than that of comparative parallel disposition. The weak vortices are also observed for the single Bridge 2 as shown in Fig. 16(f) and (l). However, the flow patterns are quite different

from those for a parallel arrangement. Alternating vertical flows are not clearly observed. The wind velocity behind Bridge 2 is less than 1 m/s for most of the phases and the low-speed area develops freely downstream without any hindrance by Bridge 1.

The comparative investigation of PIV images demonstrates that the parallel disposition of two decks amplifies alternating “Ω” and “∩” shaped flow fields which illustrates the mechanism of the observed VIV in Bridge 2.

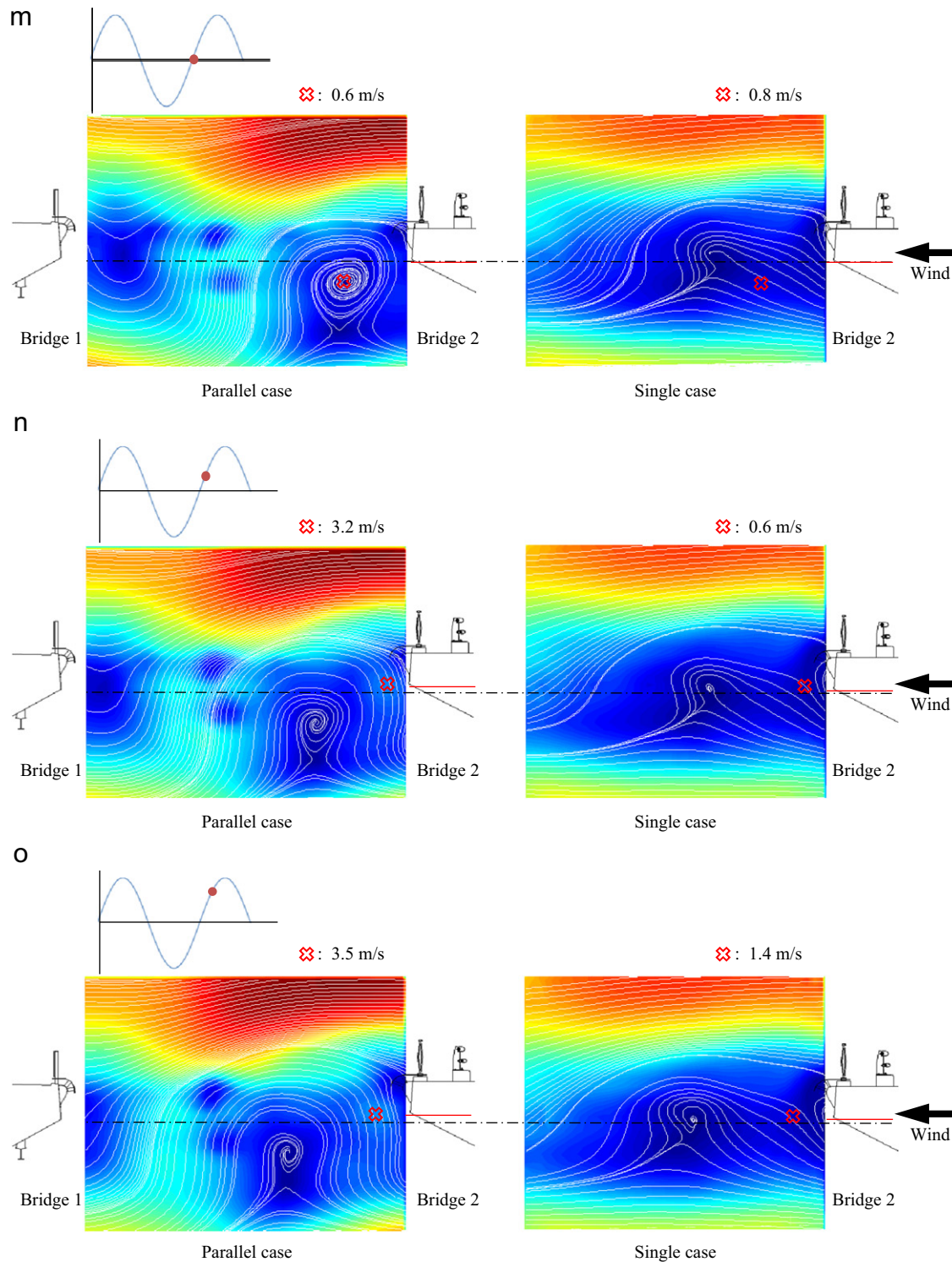


Fig. 16. Continued.

5. Conclusions

An interference effect on VIV was observed in an actual twin cable-stayed bridge for the first time. The upstream deck experienced a steady-state vibration tuning the lock-in velocity for the first vertical mode. A series of wind tunnel tests were carried out and the observed interference VIV was successfully reproduced. Through comparative wind tunnel tests for a single disposition of the bridge, it was found that the VIV of the upstream deck is

amplified due to the interference effect between parallel decks of the twin cable-stayed bridge. Some attempted modifications using aerodynamic additives were not effective in reducing the interference VIV. However, a gradual increase in structural damping was found to be effective in mitigating the amplitude of the interference vibration. The damping ratio for the first vertical mode of the upstream bridge was estimated to be 0.25–0.28% from an operational modal analysis method and this unexpected low level of mechanical damping likely explains one of the possible reasons for

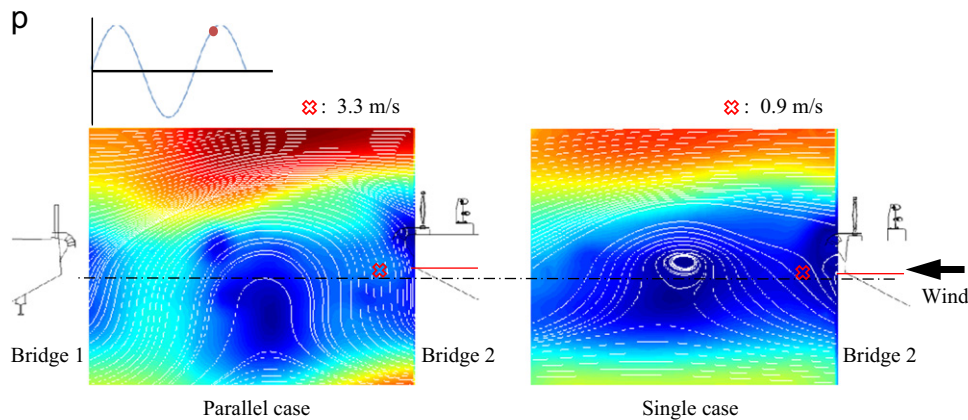


Fig. 16. Continued.

the observed vibration. Through PIV tests, alternating eddies were observed at the top and bottom corners of the upstream deck in phase with the deck vibration. The comparative investigation of PIV images demonstrates that the parallel disposition amplifies alternating “Ω” and “∩” shaped flow fields between two decks. These alternating downward and upward flows excited the upstream deck at the lock-in wind velocity.

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References

- Grillaud, G., Chauvin, A., Biétry, J., 1992. Comportement dynamique d'un pont à haubans dans une turbulence de sillage. *Journal of Wind Engineering and Industrial Aerodynamics* 41–44, 1181–1189, in French.
- Honda, A., Shiraishi, N., Matsumoto, M., Fuse, Y., Sumi, K., Sasaki, N., 1993. Aerodynamic stability of Kansai International Airport access bridge. *Journal of Wind Engineering and Industrial Aerodynamics* 49, 533–542.
- Hussain, A.K.M.F., Reynolds, W.C., 1970. The mechanics of an organized wave in turbulent shear flow. *Journal of Fluid Mechanics* 41 (2), 241–258.
- Hyundai Engineering and Construction (HDEC), 2000a. Report on the Wind Tunnel Tests for Second Jindo Bridge (in Korean).
- Hyundai Engineering and Construction (HDEC), 2000b. Report on the Monitoring of Jindo Bridge (in Korean).
- James III, G.H., Carne, T.G., Lauffer, J.P., 1993. The Natural Excitation Technique (NExT) for Modal Parameter Extraction From Operating Wind Turbines (No. SAND-92-1666). Sandia National Labs., Albuquerque, NM.
- Jang, S., Jo, H., Cho, S., Mechitov, K., Rice, J.A., Sim, S., Jung, H., Yun, C., Spencer Jr., B.F., Agha, G., 2010. Structural health monitoring of a cable-stayed bridge using smart sensor technology: deployment and evaluation. *Smart Structures and Systems* 6 (5–6), 439–459.
- Juang, J.N., Phan, M.Q., 2001. *Identification and Control of Mechanical Systems*. Cambridge University Press.
- Kim, J., 2002. Large Eddy Simulation of the Turbulent Flow Around a Square Prism. Ph.D. Dissertation. The University of Western Ontario, Canada.
- Kimura, K., Shima, K., Sano, K., Kubo, Y., Kato, K., Ukon, H., 2008. Effects of separation distance on wind-induced response of parallel box girders. *Journal of Wind Engineering and Industrial Aerodynamics* 96, 954–962.
- Korea Meteorological Administration (KMA), 2012. Annual Climatological Report 2011. KMA, Seoul, Korea.
- Korean Society of Civil Engineers (KSCE), 2006. Design Guidelines For Steel Cable-Supported Bridges, Korea.
- Larsen, S.V., Astiz, M.A., Larose, G.L., 2000a. Aerodynamic interference between two closely spaced cable supported bridges. In: *Proceedings of the 4th International Colloquium on Bluff Body Aerodynamics and Applications*, Bochum, pp. 33–37.
- Larsen, A., Esdahl, S., Andersen, J.E., Vejrum, T., 2000b. Storebælt suspension bridge—vortex shedding excitation and mitigation by guide vanes. *Journal of Wind Engineering and Industrial Aerodynamics* 88, 283–296.
- Meng, X.L., Zhu, L.D., Guo, Z.S., 2011. Aerodynamic interference effects and mitigation measures on vortex induced vibrations of two adjacent parallel bridges. *Frontiers of Architecture and Civil Engineering in China* 5 (4), 510–517.
- Siringoringo, D.M., Fujino, Y., 2008. System identification of suspension bridge from ambient vibration response. *Engineering Structures* 30, 462–477.
- Spencer Jr., B.F., Cho, S., Sim, S., 2011. Wireless monitoring of civil infrastructure comes of age. *Structure Magazine* 2011, 12–16.